

Square a Sorted Array

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Highest Square Index = 4

-1	2	2	5	7
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Left index: 0

Right index: 4

Left square: 0

Right square: 0

0	0	0	0	0
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Highest Square Index = 4

-1	2	2	5	7
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Left index: **0**

Right index: **4**

Left square: **1**

Right square: **0**

0	0	0	0	0
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Highest Square Index = 4

-1	2	2	5	7
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Left index: **0**

Right index: **4**

Left square: **1**

Right square: **49**

0	0	0	0	0
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Highest Square Index = 4

-1	2	2	5	7
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Put larger square into **highest square index**

Left square: **1**

Right square: **49**

0	0	0	0	49
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Highest Square Index = 4

-1	2	2	5	7
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Left index: **0**

Right index: **3**

Increment / decrement that index

0	0	0	0	49
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Highest Square Index = 3

-1	2	2	5	7
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Left index: **0**

Right index: **3**

Decrement highest square index

0	0	0	0	49
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Highest Square Index = 3

-1	2	2	5	7
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Left index: **0**

Right index: **3**

Left square: **1**

Right square: **49**

0	0	0	0	49
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Highest Square Index = 3

-1	2	2	5	7
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Left index: **0**

Right index: **3**

Left square: **1**

Right square: **25**

0	0	0	0	49
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Highest Square Index = 3

-1	2	2	5	7
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Put larger square into **highest square index**

Left square: **1**

Right square: **25**

0	0	0	25	49
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Highest Square Index = 3

-1	2	2	5	7
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Left index: **0**

Right index: **2**

Increment / decrement that index

0	0	0	25	49
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Highest Square Index = 2

-1	2	2	5	7
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Left index: **0**

Right index: **2**

Decrement highest square index

0	0	0	25	49
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Highest Square Index = 2

-1	2	2	5	7
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Left index: **0**

Right index: **2**

Left square: **1**

Right square: **25**

0	0	0	25	49
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Highest Square Index = 2

-1	2	2	5	7
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Left index: **0**

Right index: **2**

Left square: **1**

Right square: **4**

0	0	0	25	49
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Highest Square Index = 1

-1	2	2	5	7
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Left index: **0**

Right index: **1**

Left square: **1**

Right square: **4**

0	0	4	25	49
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Highest Square Index = 1

-1	2	2	5	7
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Left index: **0**

Right index: **1**

Left square: **1**

Right square: **4**

0	4	4	25	49
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Highest Square Index = 0

-1	2	2	5	7
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Left index: **0**

Right index: **0**

Left square: **1**

Right square: **4**

0	4	4	25	49
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Highest Square Index = 0

-1	2	2	5	7
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Left index: **0**

Right index: **0**

Left square: **1**

Right square: **4**

1	4	4	25	49
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-1	2	2	5	7
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1	4	4	25	49
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Time complexity **$O(N)$**

Space complexity **$O(N)$**